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Course Title: Internet of Things Lab

Project Proposal

Submitted to:

S. M. Zafrul Islam

Lecturer

Submitted by:

Member 1 : Elham Rafat Mahmud | **ID:** [1056]

Member 2: [Rudro Antony Mrong] | ID: [1059]

Member 3: [Md.Masud Rahman] | **ID:** [1064]

Department of CSE

University of Information Technology and Sciences

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AegisVault: Autonomous Dual-Tier Threat Mitigation and Active Asset Concealment IoT Security System

Domain

- Internet of Things (IoT) & Embedded Systems
- Cyber-Physical Systems (CPS) & Physical Intrusion Detection
- Edge Computing & Asynchronous Telemetry

Problem Statement

Conventional perimeter security alarms suffer from three critical flaws:

1. **Passive Vulnerability:** Traditional systems only sound an audible alarm or illuminate lights while leaving physical assets stationary and exposed, granting intruders sufficient time to complete theft before security forces respond.
2. **Predictable Egress & Decoy Exploitation:** Sophisticated intruders frequently exploit standard evacuation protocols—intentionally triggering primary alarms to prompt automatic vault drops or staff relocation into compromised secondary chambers.
3. **Hardware Vulnerability & False Alarms:** Standard laser tripwires can be easily blinded or bypassed using continuous handheld laser pointers, while single-sensor tripwires frequently cause false positives due to environmental factors (dust, ambient light fluctuations, or insects).

Proposed Solution

This project introduces **AegisVault**, an active cyber-physical IoT security system built on a deterministic Finite State Machine (FSM):

- **Dual-Tier Physical Defense:** When a perimeter breach occurs (Layer 1), the system locks down entry/exit points and mechanically retracts the primary asset pedestal into a concealed containment sub-chamber via micro-actuators. If the intruder anticipates this movement and tampers with the sub-chamber (Layer 2), a secondary fail-secure lockdown activates along with a distinct high-frequency alarm and critical escalation alerts.
- **Anti-Bypass Sensor Fusion:** The primary tripwire utilizes a frequency-modulated laser (pulsed at 500 Hz via PWM) paired with an HC-SR501 PIR motion sensor. Intrusion is confirmed only when the modulated light pattern is interrupted concurrently with volumetric thermal movement, eliminating false trips and preventing optical blinding.
- **Zero-Cost Biometric De-escalation:** Security clearance and lockdown de-escalation are achieved through hardware-agnostic smartphone WebAuthn biometric verification (Fingerprint / Face ID) communicating securely with the IoT backend over TLS, incorporating two-person integrity rules for Tier 2 clearances.

Objectives

1. Design and calibrate an anti-tamper, frequency-modulated optical tripwire grid integrated with passive infrared (PIR) sensor fusion.
2. Construct a dual-tier physical prototype with micro-servo actuation for automated door locking and asset pedestal concealment.
3. Implement an edge-level Finite State Machine (FSM) on an ESP32 microcontroller to guarantee deterministic transitions between normal, alert, lockdown, and reset states.
4. Establish real-time telemetry using MQTT over TLS to stream state updates to a responsive web monitoring dashboard and trigger instant push notifications (via Telegram Bot API).
5. Integrate edge camera snapshot capture upon Tier 2 breach escalation and maintain a tamper-evident audit log using sequential cryptographic hashing (SHA-256).

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Hardware Requirements

Component	Specification / Model	Qty	Key Role in System	Estimated Cost (BDT)
ESP32 DevKit V1	30/38 pin	1	Edge processing, Wi-Fi/MQTT telemetry, PWM modulation, and FSM logic	550 – 600 BDT
Red Laser	650nm 5V 5mW	2	Perimeter tripwire transmitter (modulated via PWM)	50 – 60 BDT
Light Dependent Resistor (LDR)	5mm	2	Break-beam detection paired with pull-down resistors	20 – 30 BDT
PIR Sensor	HC-SR501	1	Volumetric thermal motion detection for sensor fusion	90 – 120 BDT
Micro Servos	SG90 9g	2	Servo 1: Door deadbolt lock; Servo 2: Concealment drop platform	260 – 280 BDT
Micro Limit Switch	Tactile Switch	1	Layer 2 asset weight / compartment displacement detection	15 – 25 BDT
Piezo Buzzers	Active 5V	2	Distinct alert sound profiles (Pulsed for	30 – 40 BDT

Component	Specification / Model	Qty	Key Role in System	Estimated Cost (BDT)
			Tier 1, High-Hz for Tier 2)	
LEDs	High-Brightness White & Red	6	Room floodlights (strobe effect) and visual status indicators	20 – 30 BDT
Breadboard	Solderless (830 points)	1	Circuit assembly and rapid prototyping	110 – 130 BDT
Jumper Wires	M-M, M-F, F-F	1 set	Point-to-point hardware connections	80 – 100 BDT
Enclosure Materials	High-Density Foam Board & Adhesive	1	Structural scale model (vault room & hidden basement compartment)	100 – 150 BDT
Repurposed Smartphone	IP Webcam	1	Video stream capture and snapshot dispatch during Tier 2 breach	0 BDT (Existin...

Total Estimated Hardware Cost: ~1,325 – 1,565 BDT

Software and Technologies

- **Embedded Development:** C / C++ via Arduino IDE / PlatformIO with FreeRTOS task scheduling.
- **Network & Messaging Protocols:** MQTT over TLS (Port 8883) for millisecond-latency telemetry; HTTP/REST for admin controls.
- **Cloud Broker:** HiveMQ Cloud / EMQX Serverless (Free tier, encrypted).
- **Backend Framework:** FastAPI (Python) or Node.js (Express) for event routing, webhook management, and cryptographic audit hashing.
- **Database & Storage:** PostgreSQL / Supabase for event logs and authorized role verification.
- **Frontend Web Dashboard:** Next.js (React) or Tailwind CSS client displaying real-time sensor topologies, alert logs, and system state controls.
- **Notification Service:** Telegram Bot API for real-time mobile security alerts and automated intruder photo delivery.
- **Authentication:** WebAuthn (FIDO2) API for zero-trust smartphone biometric validation (Fingerprint / Face ID).

Feasibility Analysis

1. Technical Feasibility

The ESP32 features a dual-core 240 MHz Tensilica Xtensa 32-bit processor, built-in Wi-Fi (802.11 b/g/n), and multiple hardware PWM timers. It possesses ample computing overhead to execute laser frequency modulation, sensor debouncing, FSM transitions, and encrypted TLS handshakes concurrently without race conditions.

2. Economic Feasibility

By substituting heavy industrial hardware (such as NEMA stepper motors and standalone optical biometric modules) with lightweight SG90 servos and smartphone WebAuthn APIs, the complete physical and software prototype costs under 1,700 BDT. All software layers (brokers, database tiers, notification webhooks, and hosting) operate entirely on zero-cost academic tiers.

3. Operational & Environmental Feasibility

The scale diorama operates safely on a standard 5V 2A USB power supply. Modulating the optical beam ensures resistance against external ambient light interference and deliberate optical bypasses, while dual-factor authentication guarantees rapid operational recovery by authorized personnel.

Project Scope and Limitations

Project Scope

- End-to-end implementation of an autonomous two-tier intrusion containment system.
- Real-time edge-to-cloud bidirectional synchronization via secure MQTT.
- Scale physical demonstration containing functional motorized deadbolt locking and asset concealment.
- Multi-platform incident response incorporating automated snapshot delivery to authorized mobile devices.
- Cryptographically verifiable audit log ensuring tamper-evident accountability.

Limitations

- **Scale Constraints:** The physical containment mechanism is built as an academic bench-scale prototype using micro-servos and lightweight materials rather than industrial-gauge steel enclosures.
- **Network Dependency for Remote Features:** While core threat mitigation (servos, sirens, floodlights) executes autonomously offline at the edge, remote push alerts and cloud logging depend on active Wi-Fi connectivity.
- **Line-of-Sight Alignment:** The laser-LDR detection grid requires accurate physical alignment, which can drift if the scale model experiences heavy mechanical shocks.

Expected Outcome

Upon project completion, the team will deliver:

1. A fully operational physical diorama featuring an upper vault chamber, automated motorized door locking, and an active drop-concealment safe chamber.
2. Robust ESP32 firmware running a deterministic Finite State Machine resilient to laser blinding and transient sensor noise.
3. A functional live web dashboard providing real-time vault telemetry, active beam status, and mobile WebAuthn biometric disarming.
4. An automated emergency notification pipeline that pushes immediate breach alerts and edge camera intruder snapshots directly to Telegram within 1.5 seconds of a security trigger.
5. A formal engineering project report and presentation deck verifying system benchmarks, sensor response latency, and system reliability.